Metal alloys

23 May 2025 23:41

Pure Metals → Ductile / Soft
→ Low lattice resistance

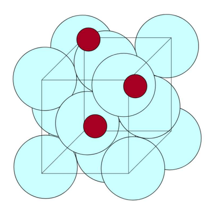
3 Methods to harden metals:

• Solid solution strengthening:

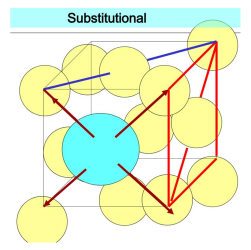
- Can harden a metal by alloying it (making impure)
- Can dissolve impurities in a solid crystal – just like in a liquid

> • E.g.: - Copper can dissolve up to 30% Zn
> - Copper and Zn atoms not same size

- Dissolving one element in solid crystal of another implies:
  - → Solute fits into spaces between atoms of crystal (Interstitial alloy)



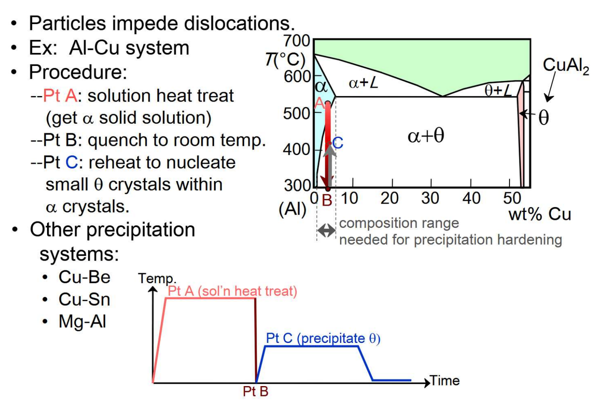
  - → Solute replaces atoms of solvent crystal (Substitutional alloy)

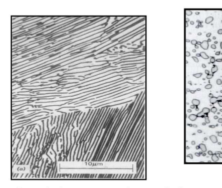


> - In each case, the lattice is distorted.
> - Introduces stresses and roughens the slip plane
> - Makes it harder for dislocations to move.

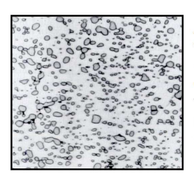
• Precipitate / dispersion hardening:

- If solution is saturated, change in temp. may result in solute precipitating out of solution.
- Happens in both liquid and solid solutions





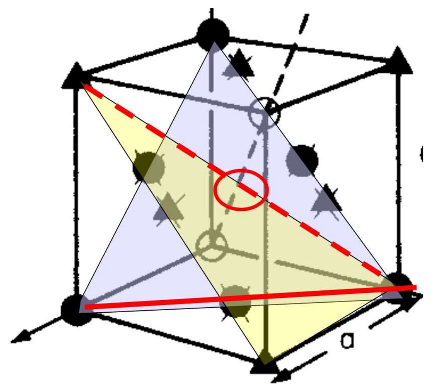
Steel is strengthened by Iron Carbide Precipitates.



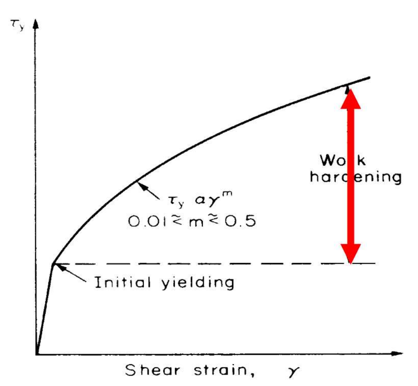
An alloy of aluminium with 4% copper (Duralumin) can be cooled to produce a fine dispersion of fine precipitates of hard $CuAl_2$

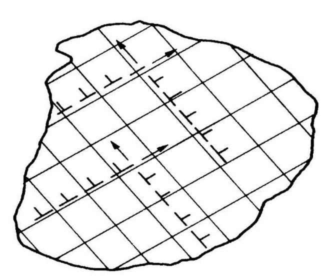
• Work hardening:

- When crystals yield, dislocations move
- Most crystals have several slip planes; FCC crystal has slip planes {1, 1, 1}



Dislocations moving on the different {111} planes will interact, intersect and obstruct each other – raising the stresses required to continue the movement. The dislocations build up and accumulate in the crystal





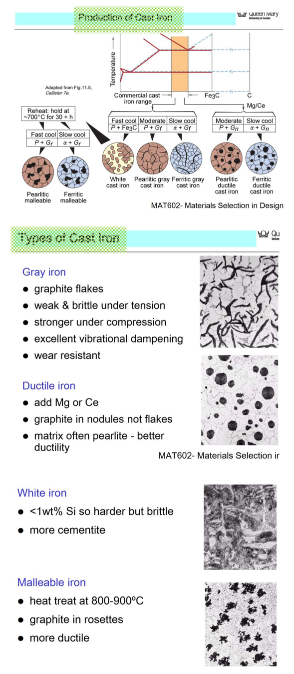
The result is work hardening – the stress required to continue plastic deformation increases after the initial yield.

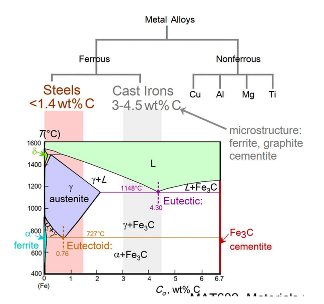
Metal types:

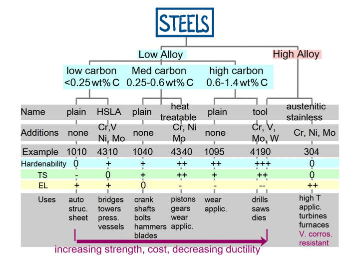
Ferrous alloys

- Iron always a prime constituent

> e.g.
> – low carbon steel, high carbon steel, stainless steel, cast iron, grey iron, white iron.

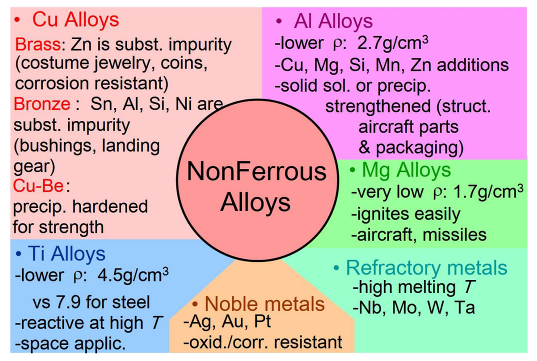






Non Ferrous Alloys:

- Copper alloys
  - Brasses (Copper & Zinc)
  - Bronzes (Copper & tin, aluminium, silicon, nickel)
- Aluminium alloys
  - → Aluminium - lithium alloys used in aircraft structures
  - → Aluminium - Copper alloys used in Chemical + food industries
- Titanium alloys
- Magnesium alloys



Process Considerations:

- Energy cost taking metal to melt temp. is too high

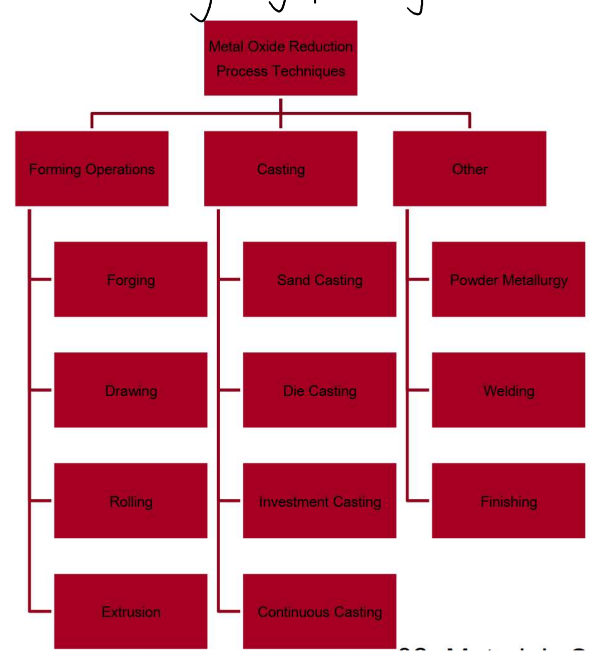
∴ Desirable to form or shape during initial transformation from melt

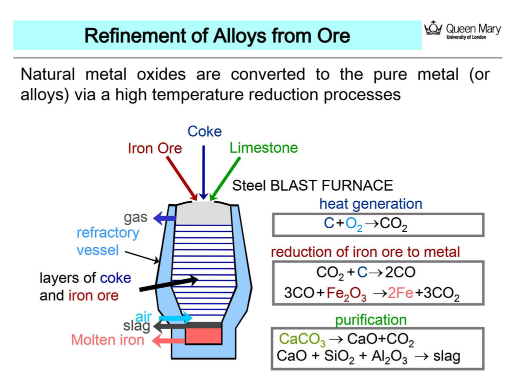
Shaping operations at melt temps require:

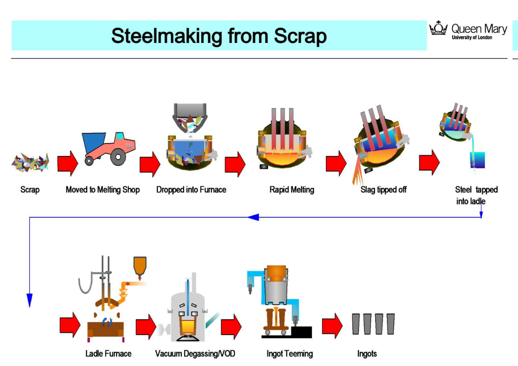
- Expensive moulds or tools made from exotic metals or materials, difficult to form accurately.

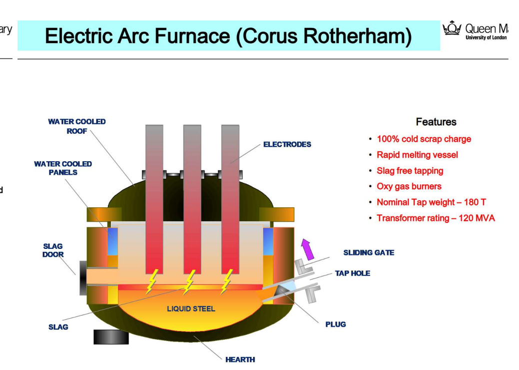
- Cooling from high temps results in significant shrinkage impacting dimensional tolerances
  ↓ Creates air gap which insulates mould
- Work hardening limits extent of plastic deformation that can occur in a single step.
- Extent of plastic forming is limited
- Metals can't be easily pressed around corners
- Most presses operate in one direction, however liquids flow in all directions
- Phase changes occur on both heating and cooling

Primary alloy processing:









Casting – mould filled with metal

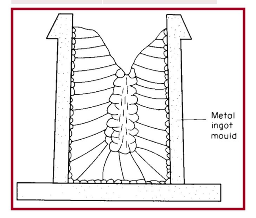
- Metal melted in furnace (alloying elements added), then cast in a mould
- Most common / cheapest method
- Good production of shapes
- Weaker products / internal defects
- Good option for brittle materials

Ingot Casting:

- Processes depend on material produced
- Mould type depends on material, hence melt temp. →

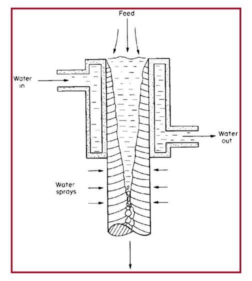
Metal	Melt temp/°C
Tungsten	3400
Silica	1900
Platinum	1770
Iron	1536
Nickel	1453
Mild steel	1413
Gold	1063
Copper	1083
Aluminium	660
Magnesium	650
Zinc	420
Lead	327

- Moulds usually simple with rounded corners to allow ease of removal.
- Metals have a grain structure
- Shrinkage → Contraction cavity at top of ingot.
- Impurities first expelled from solid and concentrated in last melt zones
- Gas bubbles may form from dissolved nitrogen released on cooling



Continuous Casting:

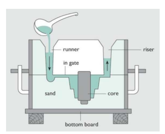
- Many problems eliminated if material is converted into shapes directly during continuous casting process to make bar or sheet
- Contraction cavities don't form as material is continuously replenished.
- Segregation reduced because columnar grains grow over smaller distances.
- Work required to convert product into rod or sheet is less because starting block is smaller

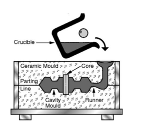


3 Approaches:

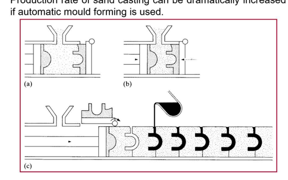
Permanent patterns (Sand casting)

- Pattern / model of part used to make impression in expendible mould
- Mould destroyed upon each use, pattern reused



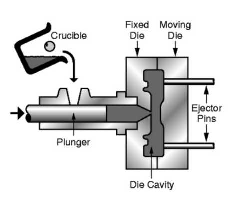


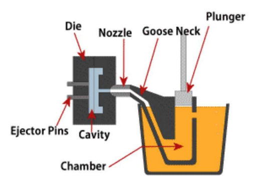
Production rate of sand casting can be dramatically increased if automatic mould forming is used.



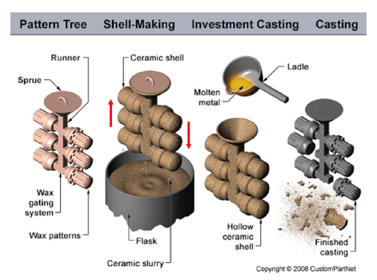
Permanent patterns (Die casting):

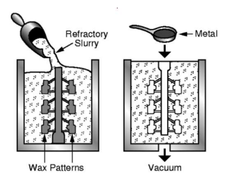
- Mould reused; part released by opening mould rather than destroying it
- Moulds made accurately from materials that can withstand high temps.
- Restricts process to low melt temp. alloys
- Moulds are expensive
- To be used economically, requires:
  - Long production runs
  - or, small components





Expendible mould and pattern (Investment casting):





Forming:

- Shape changing operations; Convert shape via plastic deformation

Why use forming at all if we have casting available?

- There are defects in castings that for a cost can be reduced.
- Geometry - Long (one dimension significantly greater than the others) shapes such as tubes are difficult to cast.
- Microstructure - Best castings come from eutectic alloys and this is not always desirable.
- Inhomogeneity - common in castings but plastic deformation (the basis for forming) reduces inhomogeneity.
- Refractory materials - Some materials are difficult to process as liquid due to high melt temperatures - usually processed as powders with some associated forming operations.

Classification:

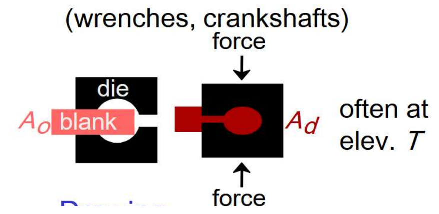
- Continuous = Rolling
- Intermittent = Forging

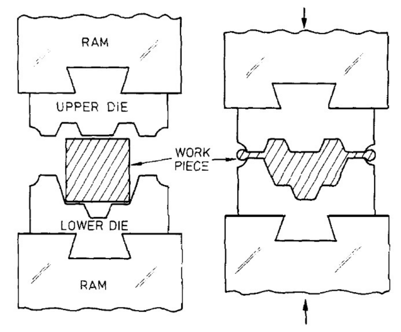
} – Bulk deformation
– Triaxial stresses
– Starting material → high V. to S.A.

Forging (Hammering, Stamping):

- Hot forging → bigger shape change, but poor surface & tolerance because of oxidation & warpage.
- Cold forging → greater precision & finish, but forging pressures are higher and deformations limited by work hardening.

(wrenches, crankshafts)



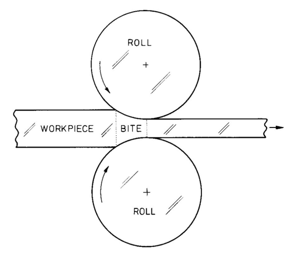


- Formability = min radius to which a sheet can be bent.
  - → Sometimes expressed in terms of sheet thickness t (1 = good ; 4 = average)
  - → Determines amount sheet can be stretched or drawn without necking and failing
- Radii best made as large as possible; never less than sheet thickness
- Forming limit diagrams give more precise info. – shows combination of principle strains in plane of sheet that cause failure
  - → Parts designed so strains don't exceed this limit

Rolling (Hot and Cold):

Cold rolling: ($T < 0.35\,T_M$)

- Exploits work hardening to increase strength of final product, at penalty of higher forming pressures.

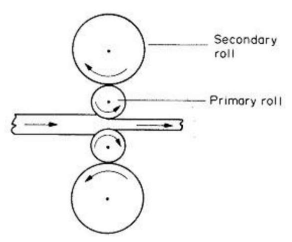


Hot rolling

- Lower work hardening + Lower pressures

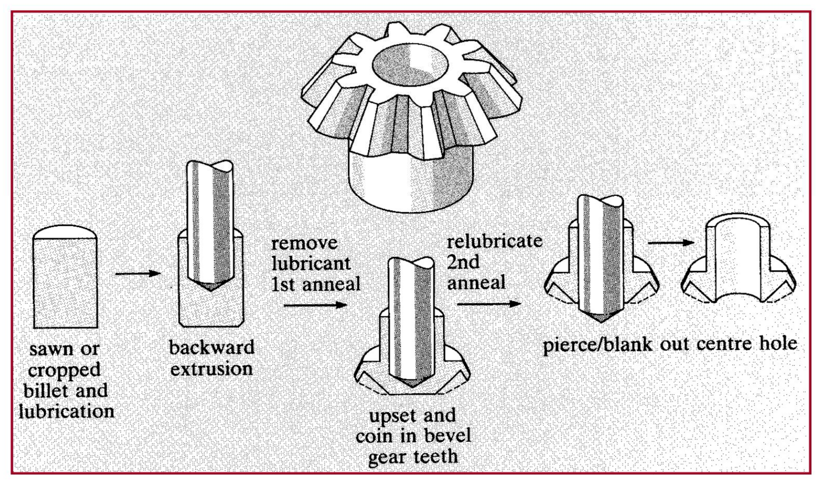
Roller geometry:

- Primary roller should always be as small as possible
- Secondary roller (if present) can be bigger



FORGING OF A GEAR

Queen Mary University of London



Material Considerations

- Real metals exhibit work hardening; limits extent of deformation possible, also increases hardness of material.
- Basis for difference between cold and hot working
  - ↳ Stiff, hard & strong
- However, forces needed to deform metal are high; resulting in in-built stresses and corrosion during service

TRI-AXIAL COMPRESSION

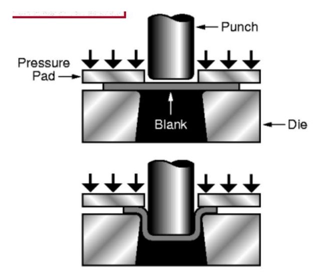
Queen Mary University of London

- **Q. Why do forming operations usually involve tri-axial compression?**

- Hydrostatic compression (tri-axial with balanced σ_1, σ_2, σ_3) involves no shear but can result in overall volume change.
- Mismatch in compressive stresses introduces shear stresses on some planes - these result in plastic deformation.
- Key point is that the hydrostatic component of the pressure ensures that defects do not propagate as tensile stresses would lead to crack growth, rupture or tearing.

Drawing: (Usually only biaxial forces)

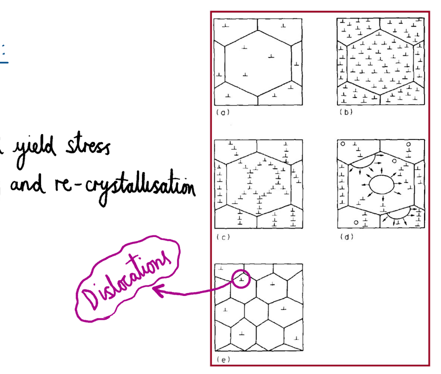
- Blanking – cutting via shear forces punch and die
- Deep drawing – bending, reverse bending
- – No change in volume ∴ side walls may increase in thickness
- – Deformation limited due to side walls forces which will ultimately fracture



Hot working:

Can exhibit:

→ reduced yield stress
→ recovery and re-crystallisation

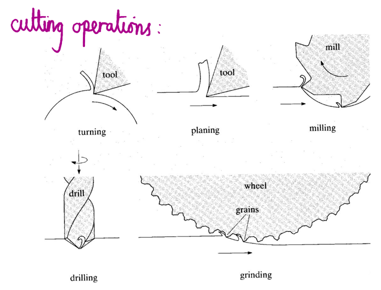


Machining:

- Secondary process, used if:
  - – Introduction of holes, slots, etc; difficult via primary process
  - – To improve dimensional tolerances or surface texture
- Sometimes a primary process (CNC, milling, grinding)
- Tooling costs reduced; if material is cheap, then wastage tolerated

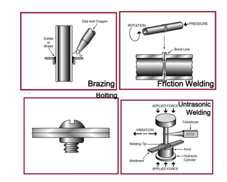
Cutting:

- Cutting tool harder than work material, and work piece ductile
- Single point cutting operations:
  - – turning
  - – Planing
  - – Milling
  - – Drilling
  - – Grinding



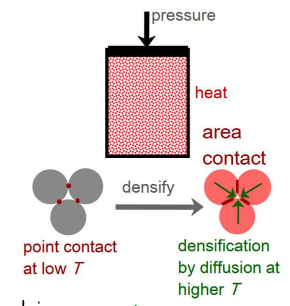
Joining:

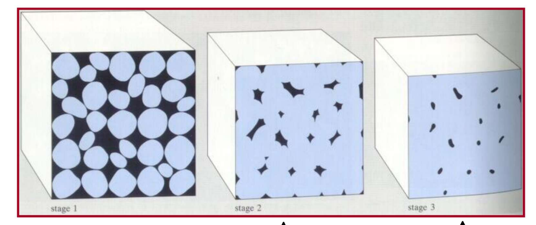
- Welding
- Fastening
- Adhesive bonding
- Bolts and screws (facilitate disassembly)



Powder metallurgy:

- Liquid metal changed to powder via atomisation process.
- Can be compacted into shapes & sintered
- Used with high melt temp materials
- Processes near net shape, reducing finishing operations
- Good if hard materials like carbides & nickel super-alloys being processed
- Unusual combos of metals / alloys possible
- Unusual / Unstable microstructures (rapidly solidified alloy powders)



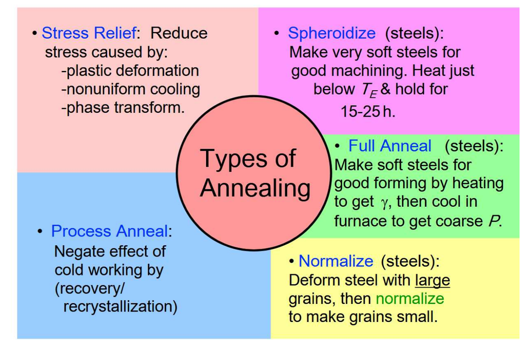


Compacting / Pressure forcing particles together

Heating up; thermodynamic process causes gaps to shrink to reduce surface energy

Thermal Processing of Metals

Annealing: Heat to T_{anneal}, then cool slowly.



Heat Treatments

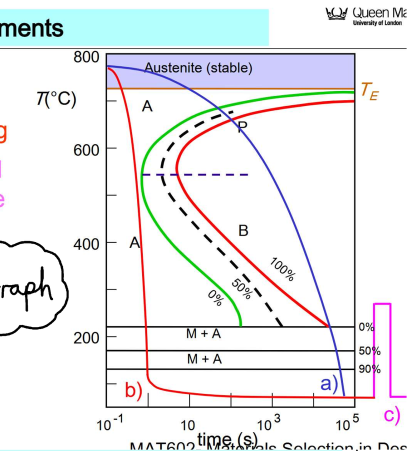
Queen Mary University of London

a) Annealing

b) Quenching

c) Tempered Martensite

TTT Graph

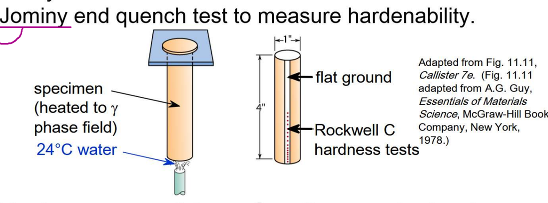


Hardenability--Steels

Queen Mary University of London

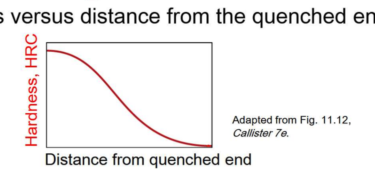
- Ability to form martensite
- Jominy end quench test to measure hardenability.

Cool by squirting water on one end that end cools v. fast. As you go down cooling end you see it cools less quickly. When machined along length, see different temps, ∴ different heat treatments taking place.



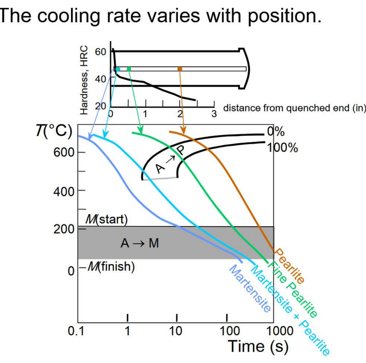
Adapted from Fig. 11.11, *Callister 7e.* (Fig. 11.11 adapted from A.G. Guy, *Essentials of Materials Science*, McGraw-Hill Book Company, New York, 1978.)

- Hardness versus distance from the quenched end.



Adapted from Fig. 11.12, *Callister 7e.*

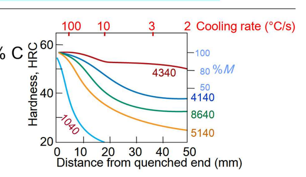
- The cooling rate varies with position.



Hardenability versus Alloy Composition

Queen Mary University of London

- Jominy end quench results, C = 0.4 wt% C



- "Alloy Steels" (4140, 4340, 5140, 8640)
  - --contain Ni, Cr, Mo (0.2 to 2wt%)
  - --these elements shift the "nose".
  - --martensite is easier to form.